VOLUME 4C — QUESTION BANK: LOGICAL REASONING (100 Questions)

> Priority: LOW–MEDIUM for DAT 2027. Many items use design or craft contexts, so practising them also revises GK.
> Sections: A. Number & letter series (1–15) · B. Analogies (16–30) · C. Classification (31–42) · D. Coding–decoding (43–52) · E. Syllogisms & deduction (53–65) · F. Statements, assumptions, cause–effect (66–78) · G. Arrangements, ranking, directions & puzzles (79–100)

Format: Question → **Answer** → explanation → *Concept* · *Diff* · *Why NID* · *Mistake*.

A. Number & Letter Series (1–15)

LR-1. 5, 11, 23, 47, 95, ?
(a) 190 (b) 191 (c) 189 (d) 193
Ans: (b). Each term is ×2 + 1: 95 × 2 + 1 = 191. *Concept:* multiplicative series · *Diff:* E · *Why NID:* logic warm-up · *Mistake:* answering 190 by forgetting the +1.

LR-2. 1, 4, 9, 16, 25, ?
(a) 30 (b) 36 (c) 49 (d) 35
Ans: (b). Squares: $6^2 = 36$. *Concept:* squares · *Diff:* E · *Why NID:* pattern recognition · *Mistake:* none.

LR-3. 2, 3, 5, 7, 11, 13, ?
(a) 15 (b) 17 (c) 19 (d) 16
Ans: (b). Primes. *Concept:* primes · *Diff:* E · *Why NID:* pattern · *Mistake:* answering 15 (15 = 3 × 5).

LR-4. 3, 6, 18, 72, ?
(a) 144 (b) 288 (c) 360 (d) 216
Ans: (c). The multipliers are ×2, ×3, ×4, so the next is ×5: 72 × 5 = 360. *Concept:* increasing multiplier · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 288 (×4 again).

LR-5. 1, 8, 27, 64, ?
(a) 100 (b) 125 (c) 81 (d) 216
Ans: (b). Cubes: $5^3 = 125$. *Concept:* cubes · *Diff:* E · *Why NID:* volume intuition · *Mistake:* none.

LR-6. 10, 9, 7, 4, 0, ?
(a) −4 (b) −5 (c) −6 (d) −3
Ans: (b). The differences are −1, −2, −3, −4, so the next is −5: 0 − 5 = −5. *Concept:* difference series · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-7. 2, 5, 10, 17, 26, ?
(a) 35 (b) 36 (c) 37 (d) 39
Ans: (c). Each term is $n^2 + 1$: $6^2 + 1 = 37$. *Concept:* squares + 1 · *Diff:* M · *Why NID:* pattern · *Mistake:* none.

LR-8. A, C, F, J, O, ?
(a) T (b) U (c) S (d) V
Ans: (b). The gaps grow by one: +2, +3, +4, +5, then +6. O(15) + 6 = 21 = U. *Concept:* letter series · *Diff:* M · *Why NID:* pattern · *Mistake:* miscounting the letter positions.

LR-9. Z, X, V, T, R, ?
(a) Q (b) P (c) O (d) S
Ans: (b). −2 each step: R(18) − 2 = 16 = P. *Concept:* backward letters · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-10. AZ, BY, CX, DW, ?
(a) EV (b) EU (c) FV (d) EW
Ans: (a). Opposite letters (each pair sums to 27). *Concept:* opposite letters · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-11. 1, 2, 6, 24, 120, ?
(a) 600 (b) 720 (c) 240 (d) 840
Ans: (b). Factorials: 6! = 720. *Concept:* factorial series · *Diff:* M · *Why NID:* pattern · *Mistake:* none.

LR-12. 4, 6, 9, 13.5, ?
(a) 18 (b) 20.25 (c) 20 (d) 17
Ans: (b). ×1.5 each step: 13.5 × 1.5 = 20.25. *Concept:* ratio series · *Diff:* M · *Why NID:* proportional thinking · *Mistake:* none.

LR-13. 2, 4, 3, 9, 4, 16, 5, ?
(a) 20 (b) 25 (c) 36 (d) 6
Ans: (b). The terms come in pairs (n, n^2), so 5 is followed by 25. *Concept:* paired series · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 6.

LR-14. 0, 1, 1, 2, 3, 5, 8, 13, ?
(a) 18 (b) 20 (c) 21 (d) 26
Ans: (c). Fibonacci: 8 + 13 = 21. (Fibonacci numbers relate to the golden ratio and to spirals in sunflowers.) *Concept:* Fibonacci · *Diff:* E · *Why NID:* design and nature link · *Mistake:* none.

LR-15. Which term is wrong in the series: 3, 7, 15, 31, 63, 128?
(a) 15 (b) 31 (c) 63 (d) 128
Ans: (d). The rule is ×2 + 1, so 63 × 2 + 1 = 127, not 128. *Concept:* the wrong-term type · *Diff:* M · *Why NID:* error-spotting · *Mistake:* none.

B. Analogies (16–30)

LR-16. Loom : Weaver :: Wheel : ?
(a) Driver (b) Potter (c) Tailor (d) Smith
Ans: (b). Tool : craftsperson. *Concept:* tool–user · *Diff:* E · *Why NID:* the craft context · *Mistake:* answering "driver" (a car wheel is not that person's main tool of making).

LR-17. Madhubani : Bihar :: Warli : ?
(a) Gujarat (b) Maharashtra (c) Odisha (d) Rajasthan
Ans: (b). Art form : state. *Concept:* region analogy · *Diff:* E · *Why NID:* Indian art GK · *Mistake:* none.

LR-18. Pashmina : Goat :: Silk : ?
(a) Sheep (b) Silkworm (c) Cotton plant (d) Spider
Ans: (b). Fibre : source. *Concept:* material source · *Diff:* E · *Why NID:* textile awareness · *Mistake:* none.

LR-19. Typeface : Font :: Family : ?
(a) Member (b) House (c) Tree (d) Name
Ans: (a). A font is one member of a typeface family. *Concept:* whole–part · *Diff:* M · *Why NID:* typography concepts · *Mistake:* none.

LR-20. Architect : Blueprint :: Composer : ?
(a) Orchestra (b) Score (c) Song (d) Stage
Ans: (b). Creator : planning document. *Concept:* creator–document · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering "song" (the product rather than the plan).

LR-21. Taj Mahal : Shah Jahan :: Qutub Minar : ?
(a) Akbar (b) Qutb-ud-din Aibak (c) Humayun (d) Ashoka
Ans: (b). Monument : the ruler who commissioned or began it. *Concept:* history · *Diff:* E · *Why NID:* architecture GK · *Mistake:* none.

LR-22. Pixel : Image :: Brick : ?
(a) Cement (b) Wall (c) Clay (d) Kiln
Ans: (b). Unit : whole. *Concept:* part–whole · *Diff:* E · *Why NID:* digital + material thinking · *Mistake:* answering "clay" (the material, not the whole).

LR-23. Bauhaus : Gropius :: De Stijl : ?
(a) Picasso (b) Mondrian (c) Dalí (d) Warhol
Ans: (b). Movement : key figure. (Theo van Doesburg was a co-founder of De Stijl; Mondrian is its most famous member.) *Concept:* design history · *Diff:* M · *Why NID:* design GK · *Mistake:* none.

LR-24. Kiln : Pottery :: Oven : ?
(a) Bread (b) Fire (c) Flour (d) Chef
Ans: (a). The heating device and what it bakes or fires. *Concept:* process–product · *Diff:* E · *Why NID:* reasoning · *Mistake:* none.

LR-25. Sculptor : Chisel :: Painter : ?
(a) Canvas (b) Brush (c) Easel (d) Colour
Ans: (b). Maker : main active tool. *Concept:* tool–user · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering canvas (the surface).

LR-26. Red : Stop :: Green : ?
(a) Wait (b) Go (c) Slow (d) Danger
Ans: (b). Signal colour : meaning. *Concept:* colour semantics · *Diff:* E · *Why NID:* signage · *Mistake:* none.

LR-27. Muga : Assam :: Paithani : ?
(a) Gujarat (b) Maharashtra (c) Karnataka (d) Tamil Nadu
Ans: (b). *Concept:* textile region · *Diff:* E · *Why NID:* textile GK · *Mistake:* none.

LR-28. Sketch : Drawing :: Draft : ?
(a) Essay (b) Pen (c) Paper (d) Reader
Ans: (a). Preliminary version : final work. *Concept:* stage relationship · *Diff:* M · *Why NID:* process thinking · *Mistake:* none.

LR-29. Cube : 6 :: Tetrahedron : ?
(a) 3 (b) 4 (c) 5 (d) 8
Ans: (b). Solid : number of faces. *Concept:* geometry · *Diff:* E · *Why NID:* form knowledge · *Mistake:* none.

LR-30. Helvetica : Sans serif :: Garamond : ?
(a) Script (b) Serif (c) Display (d) Monospace
Ans: (b). *Concept:* type classification · *Diff:* E · *Why NID:* typography · *Mistake:* none.

C. Classification / Odd One Out (31–42)

LR-31. Odd one out: Kanjeevaram, Banarasi, Paithani, Madhubani
Ans: Madhubani. It is a painting tradition; the others are woven saris. *Concept:* category · *Diff:* E · *Why NID:* craft GK · *Mistake:* none.

LR-32. Odd one out: Le Corbusier, Louis Kahn, Charles Correa, Raja Ravi Varma
Ans: Raja Ravi Varma. He was a painter; the others were architects. *Concept:* profession · *Diff:* E · *Why NID:* GK · *Mistake:* none.

LR-33. Odd one out: Futura, Helvetica, Times New Roman, Arial
Ans: Times New Roman. It is a serif; the others are sans serif. *Concept:* typography · *Diff:* E · *Why NID:* type · *Mistake:* none.

LR-34. Odd one out: Konark, Khajuraho, Lingaraja, Meenakshi
Ans: Meenakshi. It is Dravida style; the other three are Nagara-style temples. *Concept:* temple styles · *Diff:* M · *Why NID:* architecture · *Mistake:* none.

LR-35. Odd one out: Red, Yellow, Blue, Green (RYB painting primaries)
Ans: Green. It is a secondary colour in the RYB model. *Concept:* colour theory · *Diff:* E · *Why NID:* colour · *Mistake:* thinking in RGB, where green is a primary.

LR-36. Odd one out: Bandhani, Leheriya, Ikat, Chikankari
Ans: Chikankari. It is embroidery; the others are resist-dyeing techniques. *Concept:* textile process · *Diff:* M · *Why NID:* textile depth · *Mistake:* none.

LR-37. Odd one out: Picasso, Braque, Dalí, Gris
Ans: Dalí. He was a Surrealist; the others were Cubists. *Concept:* art movements · *Diff:* M · *Why NID:* art history · *Mistake:* none.

LR-38. Odd one out: Square, Rhombus, Trapezium, Cylinder
Ans: Cylinder. It is 3D; the others are 2D quadrilaterals. *Concept:* geometry · *Diff:* E · *Why NID:* form · *Mistake:* none.

LR-39. Odd one out: Dhokra, Bidri, Pietra dura, Kalamkari
Ans: Kalamkari. It is a textile art; the others are metal or stone crafts. *Concept:* craft materials · *Diff:* M · *Why NID:* craft GK · *Mistake:* none.

LR-40. Odd one out: Jony Ive, Dieter Rams, Philippe Starck, Milton Glaser
Ans: Milton Glaser. He was a graphic designer; the others are product or industrial designers. *Concept:* design fields · *Diff:* M · *Why NID:* designer GK · *Mistake:* none.

LR-41. Odd one out: 121, 144, 169, 196, 210
Ans: 210. It is not a perfect square (the others are 11^2, 12^2, 13^2, 14^2). *Concept:* number property · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

LR-42. Odd one out: Sanchi Stupa, Ajanta Caves, Hawa Mahal, Ellora Caves
Ans: Hawa Mahal. It is Rajput and secular (1799); the others are ancient rock-cut or Buddhist-era monuments. *Concept:* period / type · *Diff:* M · *Why NID:* architecture GK · *Mistake:* none.

D. Coding–Decoding (43–52)

LR-43. If COLOUR is coded as DPMPVS, how is SHADE coded?
(a) TIBEF (b) TJBEF (c) RGZCD (d) TIBFE
Ans: (a). +1 per letter: S→T, H→I, A→B, D→E, E→F. *Concept:* shift code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-44. If DESIGN is written as NGISED, how is CRAFT written?
(a) TFARC (b) TFACR (c) CRAFT (d) FTARC
Ans: (a). Reversal. *Concept:* reverse code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-45. In a code, A = 1, B = 2 ... Z = 26. What is the sum of the letters in "NID"?
(a) 27 (b) 31 (c) 33 (d) 29
Ans: (a). N = 14, I = 9, D = 4; 14 + 9 + 4 = 27. *Concept:* letter values · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-46. If each letter is replaced by its opposite (A↔Z, B↔Y...), what does "ART" become?
(a) ZIG (b) ZJG (c) ZIH (d) YIG
Ans: (a). A→Z, R(18)→27 − 18 = 9 = I, T(20)→7 = G. *Concept:* opposite-letter code · *Diff:* M · *Why NID:* logic · *Mistake:* miscounting.

LR-47. If PAINT = 5 and SKETCH = 6, then CANVAS = ?
(a) 5 (b) 6 (c) 7 (d) 4
Ans: (b). The code is the number of letters. *Concept:* simple property code · *Diff:* E · *Why NID:* logic · *Mistake:* overthinking.

LR-48. If "red blue green" means "sun sky leaf", "blue yellow" means "sky light" and "green pink" means "leaf petal", what does "blue" mean?
(a) sun (b) sky (c) leaf (d) light
Ans: (b). "blue" appears in the first two sentences, and the only common word in their meanings is "sky". *Concept:* common-word decoding · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-49. In LR-48, what does "red" mean?
(a) sun (b) sky (c) leaf (d) light
Ans: (a). blue = sky (LR-48), and green = leaf (common to the first and third sentences), so red = sun. *Concept:* elimination · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-50. If FORM is coded as 6-15-18-13, what is SPACE?
(a) 19-16-1-3-5 (b) 19-15-1-3-5 (c) 18-16-1-3-5 (d) 19-16-2-3-5
Ans: (a). Letter positions: S19 P16 A1 C3 E5. *Concept:* position code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-51. If LINE is coded as KHMD (each letter −1), how is DOT coded?
(a) CNS (b) EPU (c) CNT (d) DNS
Ans: (a). *Concept:* shift code · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-52. In a code, each symbol stands for a digit: ○ = 1, △ = 3, □ = 4. A row of symbols is read as the number formed by those digits in order (so □○ = 41). What number is △□○?
(a) 8 (b) 341 (c) 12 (d) 314
Ans: (b). △ = 3, □ = 4, ○ = 1, which gives 341. *Concept:* symbol substitution · *Diff:* E · *Why NID:* reading the rule carefully · *Mistake:* adding the digits (8) when the rule says to place them side by side.

E. Syllogisms & Deduction (53–65)

For each: decide which conclusions *definitely* follow.

LR-53. All potters are artists. All artists are observers. Conclusion: All potters are observers.
(a) Follows (b) Does not follow
Ans: (a). A chain of "all" statements is valid. *Concept:* syllogism · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-54. Some designers are painters. Some painters are poets. Conclusion: Some designers are poets.
(a) Follows (b) Does not follow
Ans: (b). Two "some" premises give no certain conclusion. *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* assuming the overlap.

LR-55. All chairs are furniture. No furniture is edible. Conclusion: No chair is edible.
(a) Follows (b) Does not follow
Ans: (a). *Concept:* syllogism · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-56. No sculptor is lazy. Some students are sculptors. Conclusion: Some students are not lazy.
(a) Follows (b) Does not follow
Ans: (a). The students who are sculptors are not lazy. *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-57. All logos are symbols. Conclusion I: Some symbols are logos. Conclusion II: All symbols are logos.
(a) Only I (b) Only II (c) Both (d) Neither
Ans: (a). "All A are B" implies "some B are A", but not "all B are A". *Concept:* conversion · *Diff:* M · *Why NID:* logic · *Mistake:* answering (c).

LR-58. Some fabrics are silk. All silk is natural. Conclusion: Some fabrics are natural.
(a) Follows (b) Does not follow
Ans: (a). *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-59. All cubes are prisms. Some prisms are hollow. Conclusion: Some cubes are hollow.
(a) Follows (b) Does not follow
Ans: (b). The hollow prisms may not include any cubes. *Concept:* the undistributed middle · *Diff:* M · *Why NID:* logic · *Mistake:* assuming it follows.

LR-60. If it rains, the street floods. The street is not flooded. Conclusion: It did not rain.
(a) Follows (b) Does not follow
Ans: (a). Modus tollens: denying the consequent is valid. *Concept:* conditional logic · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

LR-61. If it rains, the street floods. The street is flooded. Conclusion: It rained.
(a) Follows (b) Does not follow
Ans: (b). Affirming the consequent is invalid; a burst pipe could have flooded the street. *Concept:* logical fallacy · *Diff:* M · *Why NID:* critical thinking · *Mistake:* assuming it follows.

LR-62. Every NID student sketches daily. Riya does not sketch daily. Conclusion: Riya is not an NID student.
(a) Follows (b) Does not follow
Ans: (a). *Concept:* contrapositive · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-63. No square is a circle. All circles are round. Conclusion: No square is round.
(a) Follows (b) Does not follow
Ans: (b). Other things (e.g. ovals) can be round, so the premises do not rule out round squares. (Real

squares aren't round, but that fact is not deducible from these premises.) *Concept:* validity vs truth · *Diff:* H · *Why NID:* rigour · *Mistake:* using outside knowledge.

LR-64. All A are B. No B is C. Conclusion I: No A is C. Conclusion II: No C is A.
(a) Only I (b) Only II (c) Both (d) Neither
Ans: (c). "No A is C" is equivalent to "no C is A". *Concept:* E-statement conversion · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-65. Some lamps are brass. No brass is plastic. Conclusion: Some lamps are not plastic.
(a) Follows (b) Does not follow
Ans: (a). The brass lamps are not plastic. *Concept:* syllogism · *Diff:* M · *Why NID:* logic · *Mistake:* none.

F. Statements, Assumptions, Inferences, Cause–Effect (66–78)

LR-66. Statement: "Install ramps at all metro stations." Implicit assumption:
(a) Some passengers need step-free access (b) Everyone uses wheelchairs (c) Stairs are illegal (d) Ramps are cheap
Ans: (a). *Concept:* assumption · *Diff:* E · *Why NID:* inclusive-design reasoning · *Mistake:* choosing an extreme option like (b).

LR-67. Statement: "Our new bottle uses 30% less plastic." Which assumption is implicit?
(a) Less plastic is desirable (b) The bottle is free (c) Plastic is banned (d) Customers hate glass
Ans: (a). *Concept:* assumption · *Diff:* E · *Why NID:* sustainability reasoning · *Mistake:* none.

LR-68. Cause: heavy rains for a week. Effect most likely:
(a) Vegetable prices rise (b) Ice-cream sales rise (c) Fewer umbrellas sold (d) Dust storms
Ans: (a). Crop damage and transport disruption push prices up. *Concept:* cause–effect · *Diff:* E · *Why NID:* everyday reasoning · *Mistake:* none.

LR-69. Statement: "Visitors at the museum doubled after audio guides in 5 Indian languages were added." Best inference:
(a) Language accessibility may have increased engagement (b) Audio guides cause all visits (c) Museums are useless without guides (d) English is not spoken
Ans: (a). This is the cautious inference. *Concept:* inference vs overgeneralisation · *Diff:* M · *Why NID:* critical thinking · *Mistake:* choosing (b).

LR-70. Statement: "Use steel tumblers instead of paper cups in the canteen." Which argument is STRONG?
(a) Steel is shiny (b) It reduces daily waste and long-term cost (c) Paper cups are colourful (d) Everyone likes steel
Ans: (b). It is relevant and substantive. *Concept:* argument strength · *Diff:* E · *Why NID:* reasoning · *Mistake:* none.

LR-71. Two events: (A) many people started cycling to work; (B) the city built protected cycle lanes. Most likely relation:
(a) A caused B (b) B caused A (c) Both from a common cause (d) Unrelated
Ans: (b). Infrastructure typically enables behaviour. (Some argue the reverse feedback exists, but the "most likely" reading is B → A.) *Concept:* causal direction · *Diff:* M · *Why NID:* urban-design reasoning · *Mistake:* none.

LR-72. Statement: "The handloom sector employs millions but weavers' incomes are low." Which course of action is MOST useful?
(a) Ban power looms entirely (b) Connect weavers directly to markets and design support (c) Stop

handloom (d) Ignore it
Ans: (b). *Concept:* course of action · *Diff:* M · *Why NID:* craft and design policy · *Mistake:* choosing an extreme action.

LR-73. Statement: "All entries must be submitted on A3 sheets." Inference:
(a) A4 entries may be rejected (b) A3 is the largest paper (c) Everyone has A3 (d) A3 is cheap
Ans: (a). *Concept:* inference from rules · *Diff:* E · *Why NID:* reading instructions (a key exam skill!) · *Mistake:* none.

LR-74. Statement: "Since the new signage was installed, fewer visitors asked for directions." The best conclusion:
(a) The signage probably improved wayfinding (b) Visitors stopped coming (c) Signs always work (d) Guards were fired
Ans: (a). *Concept:* evaluating evidence · *Diff:* E · *Why NID:* design evaluation · *Mistake:* none.

LR-75. Which statement is a FACT rather than an opinion?
(a) The Taj Mahal is the most beautiful building (b) The Taj Mahal is made mainly of white marble (c) Mughal art is better than Rajput art (d) Agra is boring
Ans: (b). *Concept:* fact vs opinion · *Diff:* E · *Why NID:* critical reading · *Mistake:* none.

LR-76. Statement: "Children in this school carry 8 kg bags." Which conclusion is NOT justified?
(a) Bags may be heavy for young children (b) All children have back pain (c) Lockers might help (d) A timetable change might reduce load
Ans: (b). It is an over-generalisation. *Concept:* conclusion validity · *Diff:* E · *Why NID:* problem framing · *Mistake:* none.

LR-77. Argument: "We should design for the average user only." Which is the strongest counter-argument?
(a) Average users don't exist in reality; designing for diverse users (5th–95th percentile and beyond) improves usability for all (b) Averages are mathematical (c) Designers like averages (d) None
Ans: (a). *Concept:* inclusive-design argument · *Diff:* M · *Why NID:* design philosophy · *Mistake:* none.

LR-78. Statement: "The poster had too much text, so few people read it." Implicit assumption:
(a) People are less likely to read text-heavy posters at a glance (b) Posters should have no text (c) Nobody reads (d) Text is illegal
Ans: (a). *Concept:* assumption · *Diff:* E · *Why NID:* visual-communication reasoning · *Mistake:* choosing the extreme (b).

G. Arrangements, Ranking, Directions & Puzzles (79–100)

LR-79. Five designers P, Q, R, S, T sit in a row facing north. Q is to the immediate right of P. R is at the left end. T is between R and P. Where is S?
(a) Right end (b) Second from left (c) Middle (d) Left end
Ans: (a). Seats 1–5 from left: R = 1. T lies between R and P, and Q is immediately right of P, so the order begins R, T, P, Q. That leaves S at seat 5, the right end. *Concept:* linear arrangement · *Diff:* M · *Why NID:* logic · *Mistake:* none.

LR-80. In a class of 40, Asha is 12th from the top. What is her rank from the bottom?
(a) 28 (b) 29 (c) 30 (d) 27
Ans: (b). 40 − 12 + 1 = 29. *Concept:* ranking · *Diff:* E · *Why NID:* logic · *Mistake:* answering 28.

LR-81. Six people A–F sit around a circular table facing the centre. A sits opposite D. B is to the immediate left of A. C sits opposite B. F is to the immediate right of D. Who sits opposite E?
(a) F (b) B (c) A (d) C

Ans: (a). Number the seats 1–6 clockwise (seen from above). For someone facing the centre, their immediate left is the next seat clockwise and their immediate right is the next seat anticlockwise. A = 1, D = 4. B is immediately left of A, so B = 2. C is opposite B, so C = 5. F is immediately right of D, so F = 3. That leaves E = 6, and seat 3 (F) is opposite seat 6. *Concept:* circular arrangement · *Diff:* H · *Why NID:* rigorous spatial logic · *Mistake:* mixing up left and right for people facing the centre.

LR-82. A walks 10 m South, turns left and walks 6 m, turns left and walks 10 m. How far and in which direction is A from the start?
(a) 6 m East (b) 6 m West (c) 10 m North (d) 16 m East
Ans: (a). Turning left from South faces East, so A walks 6 m east. Turning left from East faces North, and 10 m north returns A to the starting latitude. A ends 6 m east of the start. *Concept:* directions · *Diff:* E · *Why NID:* spatial logic · *Mistake:* turning in the wrong direction.

LR-83. If today is Wednesday, what day will it be 100 days from now?
(a) Thursday (b) Friday (c) Saturday (d) Sunday
Ans: (b). 100 mod 7 = 2, so Wednesday + 2 = Friday. *Concept:* calendar · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-84. Four friends (Ana, Ben, Cara, Dev) each like a different craft: pottery, weaving, carving, painting. Ana doesn't like pottery or painting. Ben likes weaving. Cara doesn't like carving. Who likes carving?
(a) Ana (b) Cara (c) Dev (d) Ben
Ans: (a). Ben has weaving. Ana cannot have pottery or painting, so Ana has carving. *Concept:* grid puzzle · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-85. In LR-84, Cara likes:
(a) Pottery or painting, undetermined (b) Pottery (c) Painting (d) Carving
Ans: (a). Pottery and painting remain for Cara and Dev, and no clue separates them. *Concept:* determinacy · *Diff:* M · *Why NID:* rigour · *Mistake:* forcing a unique answer.

LR-86. A clock gains 5 minutes every hour. Set correctly at 9 am, what does it show at 3 pm?
(a) 3:25 pm (b) 3:30 pm (c) 3:35 pm (d) 3:00 pm
Ans: (b). 6 hours × 5 min = 30 min ahead. *Concept:* clock logic · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-87. Books on a shelf: Red is left of Blue; Green is right of Blue; Yellow is left of Red. Which is rightmost?
(a) Red (b) Blue (c) Green (d) Yellow
Ans: (c). Order: Yellow, Red, Blue, Green. *Concept:* ordering · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-88. A, B, C, D, E are of different heights. B is taller than D but shorter than E. A is taller than E. C is the shortest. Who is the tallest?
(a) A (b) E (c) B (d) D
Ans: (a). A > E > B > D > C. *Concept:* comparison · *Diff:* E · *Why NID:* logic · *Mistake:* none.

LR-89. Pointing to a photograph, Ravi says, "She is the daughter of my grandfather's only son." How is she related to Ravi?
(a) Sister (b) Cousin (c) Mother (d) Aunt
Ans: (a). The grandfather's only son is Ravi's father, so his daughter is Ravi's sister. *Concept:* relations · *Diff:* M · *Why NID:* logic · *Mistake:* answering cousin.

LR-90. Three boxes are labelled "Pens", "Pencils" and "Mixed". All labels are wrong. You may draw one item from one box. Which box should you pick from to relabel all correctly?
(a) Pens (b) Pencils (c) Mixed (d) Any
Ans: (c). The "Mixed" box must contain only pens or only pencils, so one item reveals its contents. The other two labels then follow by elimination. *Concept:* classic logic puzzle · *Diff:* H · *Why NID:* problem-solving · *Mistake:* none.

LR-91. A designer must arrange 3 posters (A, B, C) in a row. How many arrangements are possible?
(a) 3 (b) 6 (c) 9 (d) 27
Ans: (b). 3! = 6. *Concept:* permutations · *Diff:* E · *Why NID:* layout options · *Mistake:* none.

LR-92. With 4 colours, how many ways can you paint 2 different stripes so that the stripes are different colours (the order matters: top/bottom)?
(a) 6 (b) 8 (c) 12 (d) 16
Ans: (c). 4 × 3 = 12. *Concept:* counting · *Diff:* M · *Why NID:* colour combinations · *Mistake:* answering 16 (repetition allowed) or 6 (order ignored).

LR-93. A snail climbs 3 m up a 10 m wall each day and slides back 2 m each night. On which day does it reach the top?
(a) 10th (b) 8th (c) 7th (d) 9th
Ans: (b). After 7 days and nights it is at 7 m. On day 8 it climbs 3 m, reaching 10 m, and stops before sliding back. *Concept:* classic puzzle · *Diff:* M · *Why NID:* lateral thinking · *Mistake:* answering 10.

LR-94. You have 8 identical-looking clay beads; one is heavier. What is the minimum number of weighings on a balance scale guaranteed to find it?
(a) 1 (b) 2 (c) 3 (d) 4
Ans: (b). Weigh 3 against 3. If they balance, weigh the remaining 2 against each other. If they don't, take the heavier 3 and weigh 1 against 1; if those balance, the third is heavy. *Concept:* the divide-by-three strategy · *Diff:* H · *Why NID:* problem-solving · *Mistake:* answering 3.

LR-95. How many times do a clock's hands overlap in 12 hours?
(a) 12 (b) 11 (c) 24 (d) 22
Ans: (b). *Concept:* clock logic · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 12.

LR-96. A team of 4 has to be selected from 6 students. How many ways?
(a) 15 (b) 24 (c) 30 (d) 360
Ans: (a). C(6,4) = 15. *Concept:* combinations · *Diff:* M · *Why NID:* logic · *Mistake:* using permutations.

LR-97. Lateral thinking: a man pushes his car to a hotel and tells the owner he's bankrupt. What's happening?
(a) He's playing Monopoly (b) His car broke down (c) He's lying (d) He's a hotelier
Ans: (a). In the board game, landing on a property with a hotel can bankrupt a player. *Concept:* lateral thinking · *Diff:* M · *Why NID:* creative reasoning · *Mistake:* reading it literally.

LR-98. Which is the next figure in the pattern of matchstick squares: 1 square = 4 sticks, 2 squares in a row = 7, 3 in a row = 10? How many sticks for 10 squares in a row?
(a) 30 (b) 31 (c) 40 (d) 33
Ans: (b). Sticks = 3n + 1 = 31. *Concept:* visual-numeric pattern · *Diff:* M · *Why NID:* pattern generalisation · *Mistake:* answering 40.

LR-99. At a party, everyone shakes hands with everyone else exactly once, and there are 28 handshakes. How many people attended?
(a) 7 (b) 8 (c) 9 (d) 14
Ans: (b). n(n − 1)/2 = 28, so n = 8. *Concept:* combinations · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

LR-100. Five tiles numbered 1–5 are placed in a row so that no two consecutive numbers are adjacent. Which is a valid arrangement?
(a) 1 2 3 4 5 (b) 2 4 1 3 5 (c) 1 3 2 4 5 (d) 3 4 1 2 5
Ans: (b). Check the adjacent pairs: 2-4, 4-1, 1-3, 3-5 differ by 2, 3, 2 and 2, so none are consecutive ✓. (c) contains 3-2 and 4-5; (d) contains 3-4 and 1-2. *Concept:* constraint arrangement · *Diff:* M · *Why NID:* logic · *Mistake:* not checking every pair.

NEXT: Volume 4D — Quantitative / Mathematical Aptitude (100 questions) (`04D-VOL4-QB-QUANT.md`)